

Advanced (Habanero)

- Q1.** What is the greatest common factor of 12 and 18?
(A) 2
(B) 3
(C) 6
(D) 9
(E) 12
- Q2.** Factor out the greatest common factor from $6x + 12y$
(A) $2(3x + 6y)$
(B) $3(2x + 4y)$
(C) $6(x + 2y)$
(D) $12(x + y)$
(E) $2(3x + 6y)$
- Q3.** What is the greatest common factor of 20 and 30?
(A) 5
(B) 10
(C) 2
(D) 1
(E) 10
- Q4.** Simplify the expression $4x^2 + 12x$ by factoring out the greatest common factor
(A) $4x(x + 3)$
(B) $4(x^2 + 3x)$
(C) $4x(x + 2)$
(D) $2x(2x + 6)$
(E) $2(2x^2 + 6x)$
- Q5.** Factor out the greatest common factor from $9x^2 + 27x$
(A) $3x(3x + 9)$
(B) $9(x^2 + 3x)$
(C) $9x(x + 3)$
(D) $3(3x^2 + 9x)$
(E) $9(x^2 + 3)$
- Q6.** What is the greatest common factor of 15 and 25?
(A) 1
(B) 5
(C) 3
(D) 10
(E) 15
- Q7.** Simplify the expression $6x^3 + 12x^2$ by factoring out the greatest common factor
(A) $6x^2(x + 2)$
(B) $6x^2(x + 1)$
(C) $2x(3x^2 + 6x)$
(D) $6x(x^2 + 2x)$
(E) $6x^2(x + 3)$
- Q8.** Factor out the greatest common factor from $8x^2 + 16x + 24$
(A) $8(x^2 + 2x + 3)$
(B) $4(2x^2 + 4x + 6)$
(C) $2(4x^2 + 8x + 12)$
(D) $2x(4x + 8 + 12)$
(E) $8(x^2 + 2x + 3)$

Open Ended Questions (FRQ)

- Q9.** Simplify the expression $10x^2 + 20x + 30$ by factoring out the greatest common factor. Show all your steps and explain your reasoning.
- Q10.** Factor out the greatest common factor from $12x^3 + 18x^2 + 24x$. Show all your steps and explain why you chose this method.

Chef's Tips

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- Tip 1: Encourage students to use the distributive property to check their answers
- Tip 2: Emphasize the importance of factoring out the greatest common factor
- Tip 3: Remind students to simplify expressions completely